MIL-P-19158A(SHIPS)
13 March 1962
SUPERSEDING
MIL-P-19158(SHIPS)
29 December 1955

MILITARY SPECIFICATION

PUMPS, RECIPROCATING, DIRECT-ACTING AND POWER DRIVEN

1. SCOPE

1.1 Scope. - This specification covers direct-acting steam reciprocating pumps, and power driven reciprocating pumps including primer movers.

1.2 Classification. - Reciprocating pumps shall be of the following classes:

Class R-1 - Boiler feed, high pressure, direct-acting.
Class R-2 - General service, direct acting.
Class R-3 - General service, motor driven.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball, Annular, (General Purpose).
FF-B-185 - Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187 - Bearings, Roller, Tapered.
QQ-I-652 - Iron Castings, Gray.
QQ-N-291 - Nickel-Copper-Alloy (Monel and R-Monel) Bars, Plates, Rods, Sheets, Strips, Wire, Forgings and Structural and Special Sections.
QQ-P-330 - Phosphor Bronze Bars, Plates, Rods, Sheets, Strips, Flat Wire, and Structural Special Shaped Sections.
QQ-S-763 - Steel Bars, Shapes, and Forgings - Corrosion Resisting
GGG-P-781 - Pullers and Puller Kit, Mechanical and Mechanical Puller Attachment
MIL-R-196 - Repair Parts for Internal Combustion Engines; Packaging of
MIL-B-233 - Boxes, Repair Parts
MIL-P-735 - Primer Coating, Exterior Metal (Yellow) Zinc Chromate, Alkyd Type
MIL-S-870 - Steel Castings, Molybdenum Alloy
MIL-S-890 - Steel: Forgings and Bars for Hulls, Engines and Ordnance (Heat Treated)
MIL-S-901 - Shockproof Equipment, Class HI (High-Impact) Shipboard Application, Tests for
MIL-D-963 - Drawings, Electrical Hull and Mechanical Equipment for Naval Shipboard Use
MIL-C-2174 - Controllers, Motor Starters, and Master Switches - Direct Current Naval Shipboard Use
MIL-C-2212 - Controllers Alternating Current, Naval Shipboard
MIL-V-2741 - Valves, Pump, Synthetic Rubber
MIL-P-15024 - Plates, Identification, Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment
MIL-M-15071 - Manual Service (Instruction Books) for Shipboard Electrical and Mechanical Equipment
MIL-S-15083 - Steel, Castings.
MIL-E-15090 - Enamel Equipment, Light Gray (Formula No. 11)
MIL-P-15137 - Provisioning and Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use)
MIL-P-15328 - Primer, Pretreatment (Formula No. 117 for Metals)
MIL-H-15424 - Hand Tools; Packaging of
MIL-B-16261 - Bronze, Bearing, Castings
MIL-B-16443 - Bronze, Manganese, Castings

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MIL-B-16522 - Bronze, Aluminum-Maganese: Castings
MIL-B-16540 - Bronze, Phosphor, Castings
MIL-B-16541 - Bronze, Valve, Castings
MIL-M-16576 - Metal, Gun, Castings
MIL-W-16602 - Wire, Spring Phosphor-Bronze
MIL-P-16789 - Preservation, Packaging, Packing and Marking of Pumps General, and Associated Repair Parts
MIL-S-16974 - Steel Bars, Billets, Blooms and Slabs; Carbon and Alloy (for Reforging or Other Operations Before Heat Treatment)
MIL-M-17060 - Motors; Alternating Current, Integral Horsepower (Shipboard Use)
MIL-I-17166 - Iron Castings; Nodular Graphitic (Ductile (Iron) for Shipboard Applications)
MIL-P-17286 - Propulsion and Auxiliary Steam Turbines and Gears (Including Repair Parts, Tools, Accessories and Instruments); Packaging of
MIL-M-17413 - Motors, Direct Current, Integral H. P. Naval Shipboard
MIL-P-17545 - Primer Coating, Steel Ship Maintenance, Alkyd-Red Lead Type, Formula No. 116
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-G-17859 - Gear Assembly, Speed Decreaser; and Gear Assembly, Reverse (Naval Shipboard Use)
MIL-F-20042 - Flanges, Pipe, Bronze (Silver Brazing)
MIL-F-20670 - Flanges, Pipe, Carbon Steel, 150 P. S. I., W. S. P.

STANDARDS

MILITARY

MIL-STD-10 - Surface Roughness, Waviness and Lay
MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy

DRAWINGS

BUREAU OF SHIPS

B-153 - Packings and Gaskets, Application of
B-196 - Steel Flanges for Steel Tubing, Welding Neck
B-214 - Root Connections for Attaching Piping

PUBLICATIONS

BUREAU OF SHIPS

NAVSHIPS-250-637-2 - Instructions for Torch Brazing of Ferrous and non-ferrous Piping
NAVSHIPS-250-438-8 - Inspection and Tests of Silver-Brazed Piping Systems.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATION

Handbook H28 - Screw Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.).

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

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(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Material. -

3.1.1 Material shall be as specified herein. Material not definitely specified shall be of the best quality used for the purpose in commercial practice, and shall be subjected to review and approval by the bureau or agency concerned. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product.

3.1.2 The use of materials other than those indicated will be considered when the contractor shows the necessity for the substitution and the material proposed is readily obtainable. Substitution of materials will require the specific approval of the bureau or agency concerned in each case.

3.1.3 Materials for the pumps shall be in accordance with table I, as applicable.

Table I - Materials for class R-1, R-2 and R-3 pumps.

Application	Material	Specification
Steam cylinders and valve chests	Cast steel, class B (for temperatures not exceeding 650°F.)	MIL-S-15083
	Forged steel, alloy 1 (for temperatures not exceeding 650°F.)	MIL-S-890
	Cast carbon molybdenum steel, alloy (for temperatures above 650°F.)	MIL-S-870
Steam cylinder liners and valve chest liners	High test gray iron	QQ-I-652
	Nodular iron	MIL-I-17166
Steam cylinder pistons	High test gray iron	QQ-I-652
	Nodular iron	MIL-I-17166
	Cast steel, class B	MIL-S-15083
Steam valve pistons	High test gray iron	QQ-I-652
	Nodular iron	MIL-I-17166
	Steel, carbon and alloy grade P	MIL-S-16974
Steam end piston rods and valve rods	Forged steel, alloy 1	MIL-S-890
	Nickel-copper-aluminum-alloy, 265 Brinell hardness minimum	QQ-N-286
Liquid end piston rods or plungers	Nickel-copper-aluminum-alloy, 265 Brinell hardness minimum	QQ-N-286
Liquid end piston	Gun metal	MIL-M-16576
Liquid end cylinders, covers and valve chambers	Cast steel, class B	MIL-S-15083
	Gun metal	MIL-M-16576
Liquid cylinder liners	Nickel-copper-alloy, cast	QQ-N-288
	Bronze, aluminum-manganese, castings	MIL-B-16522
Liquid end and steam end stuffing box bushings	Bronze, bearing, casting, grade I or II	MIL-B-16261
	Bronze, phosphor, castings, grade A	MIL-B-16540
	Bronze, phosphor	QQ-R-330
Stuffing box spacers, followers and glands	Gun metal	MIL-M-16576
	Bronze, valve	MIL-B-16541
Crossheads	Forged steel, alloy 1	MIL-S-890
	Cast steel, class B	MIL-S-15083
Pump valves	Bronze, manganese, castings	MIL-B-16443
	Bronze, manganese	QQ-B-666
	Bronze, phosphor, castings, grade A	MIL-B-16540
	Bronze, phosphor	QQ-P-330
	Corrosion-resisting steel, class 302 or 303	QQ-S-763

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Table I - Materials for class R-1, R-2 and R-3 pumps (Cont'd.).

Application	Material	Specification
Pump valves (Cont'd)	Nickel-copper-alloy, cast	QQ-N-288
	Valves, pump, synthetic-rubber (classes R-2 and R-3 only)	MIL-V-2741
Pump valve seats	Bronze, manganese, castings	MIL-B-16443
	Bronze, phosphor, castings, grade A	MIL-B-16540
	Nickel-copper-alloy, cast	QQ-N-288
Pump valve springs	Wire, spring, phosphor-bronze	MIL-W-16602
Pump valve studs	Bronze, phosphor	QQ-P-330
	Nickel-copper-alloy, rolled, class A or B	QQ-N-281
	Corrosion-resisting steel, class 303	QQ-S-763
Pump valve guards	Bronze, phosphor, castings, grade A	MIL-B-16540
	Nickel-copper-alloy, cast	QQ-N-288
	Gun metal	MIL-M-16576
Cylinder tie rods	Forged steel, alloy No. 1	MIL-S-890

3.2 General design. - The design and construction of the pumps shall be the lightest and most compact possible consistent with the requirements of reliability, accessibility to repair, resistance to corrosion and erosion, and economy. The principle of reliability, however, is paramount and no compromise of this principle shall be made with the other basic requirements of design. The pumps shall be of such design that they will operate over a long period of years with a minimum of wear or erosion of any operating parts. The parts subject to unavoidable wear or erosion shall be of the best material available for the purpose so as to reduce these detrimental effects to a minimum.

3.2.1 The pumps shall operate in a manner satisfactory to the bureau or agency concerned when inclined as follows:

(a) Up to 15 degrees to either side (permanently inclined).
(b) With the vessel rolling up to 45 degrees from the vertical to either side.
(c) With the vessel pitching 10 degrees up and down from the normal horizontal plane.

3.2.2 Shockproofness. -

3.2.2.1 All equipment shall be designed to withstand shock due to firing of the ship's own armament and noncontact underwater explosions of near-miss aerial bombs, torpedoes, and mines. Every attempt shall be made to design the equipment to resist shock as shown on figure 1. These shocks shall be applied at the footings of the component. Shockmounts shall not be used unless bureau or agency concerned approval of the mount and its proposed application is given.

3.2.2.2 Equipment shall be designed to resist shock by using the accelerations shown on figure 1 for determining shock forces. The determination of stresses resulting from these forces is accomplished by the application of the usual methods of analytical mechanics. The allowable stress in components due to shock shall be the yield strength (0.2 percent offset) of the material. Normal stresses in parts, such as the tensile stress in a bolt due to initial pull-up, shall not be added to the shock stresses. It is of paramount importance that the accelerations shown on figure 1 be utilized in the design of holding down bolts, supporting feet or lugs and main structural members of the equipment and that no failure or permanent deformation occur as a result of shock. Components of an assembly mounted on a common rigid base need not be designed for higher accelerations than the values for which the whole assembly is designed unless the designer has reason to believe that higher accelerations will be experienced by the components.

3.2.2.3 The design of all complete pump units shall be such that they are capable of passing the high-impact shock test specified in MIL-S-901.

3.2.2.4 Bolts designed to be stressed in shear shall be installed in holes with a minimum of clearance, as experience indicates that large clearances allow impacting and subsequent failure.

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3.2.3 In no case shall a piece of equipment be rigidly supported from more than one plane.

3.2.4 The design, construction and workmanship of the equipment shall be such that a minimum practical noise level will result under all operating conditions.

3.2.5 Pump units shall be horizontally or vertically mounted, as specified (see 6.1).

3.2.5.1 Direct-acting pumps shall be provided with adequate feet or brackets for securing to the foundation or ship's structure. Allowance shall be made for expansion of the pump due to the temperature of the steam and of the liquid pumped. To this end mounting bolt holes in the pump brackets or foundation shall be suitably slotted in the direction of expansion, or other means for expansion shall be provided, as approved by the bureau or agency concerned.

3.2.5.2 Each horizontal motor driven unit shall be mounted on a common bedplate or base. For base mounted units, the base may be either integral with the pump casing or a separate piece, and the base shall have provision for bolting the foundation.

3.2.5.3 Bedplates and bases shall be of cast steel, class CW of MIL-S-15083, or structural steel, MIL-S-19795, fabricated by welding and designed so that they shall be sufficiently rigid to permit handling shipment and installation of the unit onboard ship without disturbing the alignment of the assembled units, and such that the normal distortion, weaving or vibration of the supporting structures onboard ship cannot cause misalignment between the pumps and driving units.

3.2.5.4 Each component part of an assembled unit supported directly by the bedplate or base shall be doweled thereto to facilitate reassembly and maintenance of alignment.

3.2.5.5 All bearing and seating surfaces of bedplates, bases, brackets or pump feet shall be finish machined.

3.2.6 Steam cylinders and chests. -

3.2.6.1 Steam cylinders shall be counterbored at both ends, leaving the working bores of such length that the piston rings shall slightly override at the top and bottom when making full stroke. All cylinders shall be lined full length with liners not less than 7/32 inch thick supported by the cylinder throughout their length and firmly secured in a manner satisfactory to the bureau or agency concerned. The cylinder walls shall have sufficient thickness to allow reboring to a radial depth of 1/8 inch for the use of oversize liners.

3.2.6.2 The steam chests shall be cast separately from the cylinders and shall be secured thereto and shall be held in alignment with the cylinder in a manner satisfactory to the bureau or agency concerned. The steam chest walls shall have sufficient thickness to allow reboring to a radial depth of 1/8 inch.

3.2.6.3 Bosses or brackets shall be cast on the cylinders or cylinder covers for attachment of the rods and for securing to foundations. Jack bolts and lifting bolts shall be provided as required for convenience in disassembly.

3.2.6.4 All cylinder heads and steam chest covers shall be flanged.

3.2.6.5 The steam and exhaust connections shall be located on the steam cylinder casting or elsewhere than on the valve chest casting so as to avoid the necessity of disconnecting steam and exhaust piping when removing steam valve gear for overhaul.

3.2.6.6 The lower cylinder head and valve chest to cylinder joints shall be made tight by metal to metal ground joints, without gaskets. The top cylinder head and valve chest covers may be made up with metal gaskets in accordance with the specification listed on Drawing B-153. The lower cylinder head may be cast integral with the cylinder.

3.2.6.7 The steam cylinders of all reciprocating pumps above 6 inches bore diameter shall be fitted with 3/4-inch diameter indicator fittings and closed with screwed plugs.

3.2.6.8 All parts of steam cylinders and valve chests, where condensate can accumulate, shall be provided with drain valves and pipes of ample size.

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3.2.7 Steam pistons. -

3.2.7.1 Steam pistons shall have followers and packing ring or rings, as approved by the bureau or agency concerned, for cylinders 6 inches in diameter or less, pistons may be of box type approved by the bureau or agency concerned for the specific application, fitted with snap rings.

3.2.7.2 Followers shall be secured by collar studs, and with the nuts prevented from backing off by split pins, or as approved by the bureau or agency concerned.

3.2.7.3 The steam pistons shall be taper bored for the piston rod.

3.2.8 Steam valve gear. -

3.2.8.1 The steam valve gear shall be of a compact, simple, durable and reliable type. The main steam valve shall be of the steam thrown piston type. A flat face or other type of main steam valve not inherently balanced as regards steam pressure will not be acceptable. The movement of the main steam valve shall be controlled by the alternate admission of steam to and the release of steam to exhaust from opposite ends of the main valve operating piston cylinder; such admission of steam, or release to exhaust, shall be controlled either by a pilot valve or automatically, by the movement of the main steam piston of the pump.

3.2.8.2 The pilot valve, where used, shall be of the reciprocating piston type. The pilot valve stem or link rods shall be fitted with tappets, collars, or other means satisfactory to the bureau or agency concerned, for adjustment and regulation of the stroke of the pump. The pilot valve stem packing shall not be subjected to pressure in excess of the pump exhaust pressure.

3.2.8.3 The valve chests shall be fitted with removable liners designed in such a manner that they will serve as removable seats for the valves. The valve chest walls shall have sufficient thickness to allow reboring to a radial depth of 1/8 inch for the use of oversize liners.

3.2.9 Piston rods. -

3.2.9.1 Piston rods shall be of adequate size for the operating loads. The piston rods shall be accurately finished to a true cylindrical surface. They shall be tapered on the piston ends and shall seat on a shoulder coincident with the fit on the taper, and they shall be secured in the pistons by locked nuts or other means satisfactory to the bureau or agency concerned. They shall be threaded on the other ends for attaching to the crosshead, and the extreme end of the rods shall be notched to suit a taper pin inserted through the cross head and the notched end of the rods to prevent the rods from backing out.

3.2.9.2 The piston rods shall be provided with wrench flats adjacent to crossheads.

3.2.10 Crossheads. - Crossheads shall be of the split type.

3.2.11 Stuffing boxes and packing. -

3.2.11.1 Stuffing boxes shall be of adequate depth and so designed that when properly packed, leakage will be reduced to a minimum under all operating conditions.

3.2.11.2 Pump gland packing shall conform to the applicable specifications listed on Drawing B-153.

3.2.11.3 All stuffing box glands shall be set up by nuts threaded on gland swing bolts or studs secured in the cylinders or cylinder heads.

3.2.12 Liquid cylinders and valve chambers. -

3.2.12.1 Liquid cylinders over six inches inside diameter shall be fitted with liners not less than 7/32 inch thick supported by the cylinder throughout their length and firmly secured in a manner satisfactory to the bureau or agency concerned.

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3.2.12.2 Liquid cylinders and valve chambers shall be so designed that the pistons and valves are easily accessible and fitted for overhauling without disturbing the framing, piping or cylinders.

3.2.12.3 The covers of pump cylinders and valve chambers shall be flanged, and in all pumps when covers are removed, the openings left shall be large enough to withdraw the plungers and valves.

3.2.13 Liquid pistons. -

3.2.13.1 Liquid pistons shall be of the solid box type or body and follower type.

3.2.13.2 Followers shall be secured by collar studs, and with the nuts prevented from backing off by split pins, or as approved by the bureau or agency concerned.

3.2.13.3 The liquid piston shall be taper bored for the piston rod.

3.2.13.4 The liquid piston packing shall be in accordance with the applicable specifications listed on Drawing B-153.

3.2.14 Pump valves. -

3.2.14.1 Pump valves shall be of the disk or bevel wing type and shall be small enough to operate quietly but a sufficient number shall be provided to keep the fluid velocity through the valves at a minimum.

3.2.14.2 Disk valves shall be held concentric on their seats by studs, valve bolts or valve stems secured in a manner satisfactory to the bureau or agency concerned. The studs; bolts or stems shall also carry the valve guards which shall be secured against backing off. The valves shall be held to their seats by light helical springs.

3.2.14.3 The bevel wing type valves shall have valve guides and a valve stem cast integral with the body of the valves. The valve stems shall carry the valve guards which shall be secured against backing off. The valves shall be held to their seats by light helical springs.

3.2.15 Cradle. - A rigid cradle shall connect the steam ends and liquid ends of the pump. The cradle shall allow access to the piston rod stuffing boxes. Substantial tie-rod construction may be used.

3.2.16 Stroke index. - All pumps fitted with crossheads shall have the top and bottom of the full stroke plainly marked on adjacent fixed and moving parts. A suitable stroke indicator shall be fitted to the moving parts.

3.2.17 Motor driven units. -

3.2.17.1 The crank shafts of motor driven units shall be of the balanced type.

3.2.17.2 Connecting rods shall be bored for crosshead or wrist pins and the crank end shall be fitted with removable, split, bearing linings.

3.2.17.3 The gear trains or chain sprockets shall be designed to impart the necessary flywheel effect to the crank shaft or flywheels may be fitted, if necessary.

3.2.17.4 The motor driven pumps shall be either of the trunk-piston or crosshead type.

3.2.17.5 The complete motor driven unit shall be provided with an efficient, self-contained lubricating system. Adequate provision shall be made, by means of suitable piston-rod wipers and packing glands, for preventing lubricating oil from the crank case being carried into the cylinders.

3.2.17.6 Bearings. -

3.2.17.6.1 Radial bearings. - Radial bearings may be of the ball, roller or sleeve type.

3.2.17.6.2 Ball and roller bearings for radial loads and the installation thereof shall be in accordance with FF-B-171, FF-B-185 and FF-B-187, as applicable.

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3.2.17.6.3 Sleeve radial bearings may be of the fixed or self aligning type as approved for design and application by the bureau or agency concerned. Ample bearing surface shall be provided for each sleeve bearing.

3.2.17.7 Bearing installation. -

3.2.17.7.1 Where sleeve bearings are fitted in crankcases or elsewhere they may be of the solid bushing type secured by pins or keys. Independent bearings on brackets shall be split along the axis. All sleeve bearings shall be securely fitted to prevent axial movement.

3.2.17.7.2 For either internally or externally mounted bearings, the bearing housing shall be cast integral with the crank case or bearing bracket, or secured thereto in such a manner as to insure alignment.

3.2.17.7.3 Bearing brackets shall be either cast integral with the crank case, securely bolted thereto on a machined shoulder, or held in alignment by at least two heavy dowels and securely bolted thereto. The use of bolts alone for securing brackets is prohibited.

3.2.17.8 Bearing lubrication. - Externally mounted ball or roller bearings shall be lubricated in accordance with FF-B-171, FF-B-185 and FF-B-187, as applicable.

3.2.17.9 Bearing arrangement. - The number, type and arrangement of bearings shall be as approved by the bureau or agency concerned.

3.2.18 Steam cylinders and valve chests. - Steam cylinders and valve chests shall be insulated and lagged in a manner satisfactory to the bureau or agency concerned.

3.2.19 Air chambers. - Air chambers, when required, shall be as specified (see 6.1).

3.2.19.1 Air chambers shall not be installed on either the suction or discharge side of reciprocating boiler feed pumps on vessels having deaerating feed systems.

3.2.19.2 Air chambers when required shall be provided on the discharge side of fire, bilge and make-up feed pumps.

3.2.19.3 The volume of suction air chambers shall be not less than 1.25 times the displacement of the liquid cylinder per single stroke.

3.2.19.4 The volume of discharge air chambers shall be not less than the displacement of the liquid cylinder per single stroke.

3.2.20 Piping and valves. - All suction, discharge, steam inlet, steam exhaust and valve connections shall be flanged; the flanged connections shall be in accordance with Drawing B-196 and Specifications MIL-F-20042 and MIL-F-20670, as applicable. Root connections shall conform to Drawing B-214.

3.2.21 Threaded parts. - Threaded parts shall conform to Handbook H28.

3.2.22 Welding. - Welding shall be in accordance with MIL-STD-278.

3.2.22.1 Silver brazed joints shall be in accordance with NAVSHIPS-250-637-2. For joints 1/2 inch and above, only preinserted ring fittings shall be used.

3.2.23 Painting. - For pumps operating at a temperature of 300°F. or less, all external unmachined surfaces of ungalvanized ferrous metal parts shall be thoroughly cleaned and coated with one coat of pretreatment conforming to MIL-P-15328, and one coat of zinc chromate primer. Conforming to MIL-P-735, or one coat of red lead primer conforming to MIL-E-15090.

3.2.23.1 Painting of external surfaces of nonferrous parts of pumps will not be required.

3.3 Pump characteristics. - The general range of pump characteristics shall be in accordance with table II. Characteristics of pumps to be procured shall be specified in the individual contracts or orders.

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Table II - Pump characteristics.

Pump class	Rated capacity	Discharge pressure	Suction conditions	Range of liquid temperatures under operating conditions
	G. p. m.	P. s. i. g.	In. Hg.	°F.
R-1	75-200	350-775	Note 1	90-250
R-2	10-1400	50-350	5-25	40-100
R-3	10-100	15-100	5-15	40-100

Note 1. - Suction conditions vary from 50 p. s. i. g. positive head at 250°F. to 10 inches Hg. negative lift at 90°F.

3. 3. 1 All direct-acting pumps for general service, class R-2, shall have steam and liquid cylinders so proportioned that the pumps will deliver full rated discharge pressure with steam pressure at the inlet to the pump reduced to 80 percent of normal.

3. 3. 2 The liquids which shall be handled by class R-2 and R-3 pumps are: fresh water, sea water, brine, mixtures of sea water, oil, dirt and sediment having a maximum specific gravity of 1. 06; gasoline, lubricating oil, diesel fuel oil and boiler fuel oil.

3. 4 Motor drives. -

3. 4. 1 Crank shaft. - The pump crank shaft may be geared to or chain connected to the driving units as specified or approved in each case by the bureau or agency concerned.

3. 4. 2 Motors. - Motor drives shall be of the type and characteristics specified in the contract or order and shall conform to MIL-M-17143 for direct current motors, and to MIL-M-17060 for alternating current motors.

3. 4. 3 Reduction gears. - Where gears are used, they shall be of the spur, single or double helical, or worm type in accordance with MIL-G-17859.

3. 4. 3. 1 Reduction gears shall be independent of the pump and connected thereto by a flexible coupling. Such reduction gears may, however, be an integral part of the driving unit.

3. 4. 4 Motor rating. - The horsepower rating of each motor shall be not less than that required by the pump under maximum power conditions as determined by actual tests of the identical or similar equipment. If such data is not available the required horsepower rating of the motor shall be determined by the following formula:

$$\text{Horsepower of driver} = \frac{1.05\ (\textit{max. hp. output of the pump})}{(\text{eff. of pump})}$$

3. 4. 4. 1 The actual motor rating shall be in accordance with Navy standard motor ratings.

3. 4. 5 Controllers. - Motor controllers and pressure switches shall be in accordance with MIL-C-2174 for direct current and MIL-C-2212 for alternating current, and shall have the characteristics as specified in the contract or order.

3. 5 Identification plates and information plates. -

3. 5. 1 Identification plates and information plates shall be furnished on each pump. These plates shall be type A or B in accordance with MIL-P-15024, except that identification plates of aluminum or plastic shall not be used.

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3.5.2 Identification plates and information plates shall be firmly secured to the equipment with corrosion resistant metallic screws.

3.5.3 Pump identification plates shall contain data as follows:

(a) Manufacturer's name, model, type, size and serial number.
(b) Service application.
(c) Salient design characteristics:
    (1) Capacity in gallons per minute.
    (2) Discharge pressure.
    (3) Suction lift or suction head.
    (4) Piston speed in feet per minute.
    (5) Brake horsepower.
    (6) Test pressure.
(d) Contract number (and item number for multiple unit orders).
(e) Federal stock number.
(f) Section for inspector's stamp.

3.6 Drawings.-

3.6.1 Drawings shall be in accordance with MIL-D-963.

3.6.2 Class B drawings of the units (pump and driver as applicable) shall be furnished on each contract or order. Class B drawings shall comprise two sheets. The first sheet shall contain all the information required by MIL-D-963 for certification data. Completion and submission for final approval of the first sheet shall not be delayed by action required to complete the second sheet.

3.6.2.1 The second sheet shall contain complete performance curves. These curves shall be the performance acceptance tests curves required by 4.2.1 and 4.2.2. The curves shall be on graphs which shall have dimensions approximately but not greater than 7 by 10 inches. Each graph shall contain the data required by 4.2.3. Promptly after completion of the performance acceptance tests the second sheet shall be completed.

3.6.2.1.1 After approval of the sheet of the Class B drawing which contains the curves, four positive (white background) prints of each graph shall be forwarded to the bureau or agency concerned. Each graph shall be on a sheet which has the same dimensions as the sheets comprising the service manuals (see 3.7). Each graph shall be arranged on a sheet in such manner that all information shown thereon may be easily read after insertion, by the bureau or agency concerned, in the technical manual. Reference to these curves shall not be contained in the index or text of the manual. Delivery of the final manuals shall not be delayed by actions required to complete and deliver to the bureau or agency concerned the four sets of pages showing the performance acceptance test curves.

3.6.3 Pump master drawings shall include detail drawings of all parts, and sectional assembly drawings with lists of material comprising all parts.

3.6.3.1 Detail drawings shall contain a number of details on each sheet; details of a single pump part on small size sheets, resulting in a multiplicity of drawing sheets, shall be avoided.

3.6.3.2 The sectional assembly drawing shall contain a complete list of material with references to detail drawings. The assembly drawing shall not include couplings or brackets which may vary with different contracts or orders.

3.6.3.3 Any subassembly made up of parts which require special alignment or assembly methods which cannot be disassembled, repaired and reassembled onboard ship without the use of special tools and jigs shall be indicated as a subassembly in the list of material.

3.6.4 On detail drawings, surface roughness, waviness and lay shall be specified in accordance with requirements and symbols specified in MIL-STD-10.

3.6.5 Drawings for drivers of class R-3 pumps and associated equipment shall be in accordance with their respective specifications.

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3.6.6 The center of gravity of the complete unit shall be indicated on the outline drawing.

3.7 Service manuals.-

3.7.1 Service manuals shall be in accordance with MIL-M-15071.

3.7.2 Identification piece numbers of parts referred to in service manuals shall be identical to the identification piece numbers shown on approved master drawings.

3.7.3 Service manuals shall contain reproductions of dimensioned outline drawings of all units.

3.7.4 Preliminary service manuals submitted for approval shall include all proposed sections of the books.

3.7.5 Two copies of service manuals shall be packed with each pump unit. Additional copies shall be furnished as specified (see 6.1). If final service manuals are not available at the time pumps are shipped, then 2 copies of preliminary service manuals shall be packed with each pump unit. The final manuals shall be delivered at a later time as specified in MIL-M-15071.

3.8 Repair parts and tools.-

3.8.1 Onboard.- Repair parts shall be furnished and processed in accordance with MIL-P-15137.

3.8.2 Each set of repair parts and tools specified in the contract or order shall contain the parts listed in table III. Unless otherwise specified in the contract or order, the quantity of repair parts per set shall be based on one pump per ship.

Table III - Onboard repair parts.

Item	Name of part	Quantity per set
a	Pump valves, with guards, bolts and springs, complete	100 percent complete replacement for all installed units per ship
b	Ball bearings	
c	Steam piston rods and nuts, complete	One to four units per ship, 100 percent replacement for one pump Over four units per ship, 100 percent replacement for two pumps
d	Liquid piston rods and nuts, complete	
e	Valves stems and nuts, including valve and link rods, and tappet nuts, complete	
f	Steam piston rings for main and valve pistons, 0.010 inch diameter oversize and rough turned, complete sets	
g	Bearings, sleeve type, complete sets	
h	Bushings, including throat bushings, complete sets	
i	Main steam valve chest liners	
j	Auxiliary steam valve chest liners	
k	Main steam valve piston	
l	Discharge valve seats, complete sets 1/	
m	Suction valve seats, complete sets 1/	

1/For salt water pumps only.

3.8.3 Repair parts such as those for reduction gears; motors, and controllers, shall be furnished in accordance with the applicable component specifications or as specified (see 6.1).

3.8.4 Special tools and wrenches.-

3.8.4.1 Each set of onboard repair parts shall include a complete set of all special tools and wrenches required for maintenance of the pumps, gears and drivers. Special tools are defined as those tools not listed in the Navy Stock List of General Stores (copies of this stock list may be consulted in the office of the Government inspector.

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3.8.4.2 One set of special operating tools and wrenches shall be furnished with each pump unit, and shall include all wrenches and tools required for the operation of the complete pump units. Each set of special operating tools and wrenches shall be furnished in separate boxes. If there are no more than two such tools per set they may be attached to the pumps in a secure but readily accessible manner, rather than boxed.

3.8.4.3 Each tool or wrench shall be indelibly marked indicating the intended purpose.

3.8.4.4 Tools which are available in the Catalog of Naval Material, General Stores Section, such as common wrenches and standard pullers, will not be required. Standard complete or limited sets of pullers or parts of puller sets in accordance with GGG-P-781 shall be identified as to type and use in notes on the outline or section assembly drawings. Special tools shall be detailed, and included in the list of material.

3.8.4.5 Each box containing onboard repair parts shall contain a list entitled "List of onboard repair parts and tools". The list shall be in a format suitable for use on outline drawings and in the instruction books. It shall not be less than nominal 8-1/2 by 11 inches in size.

3.8.4.5.1 The format of the list shall include a heading and columns of data for the items listed. The heading shall include titles and applicable entries as follows:

Reproduced from drawing number _____
Number of ships _____
Application _____
Contract number _____
Manufacturer _____
Quantities are for _____ units per ship
The columns shall include:
Piece number _____
Name of part _____
Quantity _____
Drawing number _____
Federal stock number _____
Additional columns may be used as applicable

3.8.4.5.2 The list shall be so treated as to be resistant to oil, water and fading.

3.8.5 Stock.- The selection, stock numbering and quantities of repair parts for stock shall be determined and processed in accordance with MIL-P-15137.

3.8.5.1 Unless otherwise specified in the contract or order, the cognizant supply demand control point for repair parts for stock shall be Ships Parts Control Center, Mechanicsburg, Pennsylvania.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Acceptance tests.-

4.2.1 The driving unit of class R-3 pumps shall be tested as required by the applicable specifications. During these tests the pump shall be operated at sufficient load to demonstrate conformance to the suction lift and discharge pressure requirements.

4.2.2 One pump of each size and design on each contract or order, complete including the motor and controller, and reduction gear, if any, shall be given a test to determine the overall power or steam consumption of the unit under the conditions of the contract or order. This test may be conducted at the shop of the main contractor or subcontractor as arranged with the bureau or agency concerned. The test shall continue for at least two hours and shall demonstrate the ability of the pump unit to handle the liquid at the capacity, suction lift or pressure, temperature and discharge pressure specified in the contract or order.

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4.2.2.1 All tests shall be made with the pumps discharging into an air bound receiver in which the pressure is maintained constant at a pressure equal to the rated discharge pressure of the pump. Test of the pump in which the pump discharges through a throttling valve into an open tank will not be acceptable.

4.2.3 Sufficient data shall be taken during the tests specified in 4.2.1 and 4.2.2 to prepare pump characteristic curves as specified below. The curves shall be corrected in all cases to the steam or the electric *power operating conditions specified in the contract or order.* Complete operating characteristics of the pump and its driver, as specified in the contract or order, as well as the operating characteristics of the unit during the particular test run, shall be shown on each curve sheet. The capacity shall be expressed in gallons per minute, the discharge pressure in pounds per square inch gage, the suction lift in inches of mercury, the viscosity in Saybolt seconds universal (unless otherwise specified in the contract or order) and the speed in revolutions per minute or piston speed in feet per minute as applicable. Curves shall be obtained for 100, 75 and 50 percent of rated discharge pressure and shall be furnished as follows:

(a) A curve of pump speed versus capacity at constant specified discharge pressure and at constant specified maximum suction lift when handling the liquid specified in the contract or order.
(b) A curve of pump speed versus indicated horsepower.
(c) A curve of pump speed versus pounds of steam per indicated horsepower per hour.
(d) A curve of pump speed versus volumetric efficiency.
(e) A curve of pump speed versus mechanical efficiency.
(f) A curve of pump speed versus power input to the driving unit expressed in pounds of steam per hour or electrical horsepower as applicable.

4.2.4 In addition, all pumps on a contract or order shall be assembled for a performance acceptance test sufficient to demonstrate harmony of assembly, also accuracy of dimensions and alignment. Each test may be conducted at low speed and low pressure but shall include a satisfactory run of at least 30 minute duration.

4.2.5 The capacity of the pump (selected as specified in 4.2.2) may be determined by any convenient method that will insure obtaining results with probable errors not greater than 2 percent. All gages shall be calibrated before and after the tests or oftener, if necessary. The specified viscosities shall be maintained within 5 percent regardless of temperatures.

4.2.6 Reports.-

4.2.6.1 Complete performance acceptance test reports, including copies of recorded test data and characteristic curves plotted therefrom and covering the following shall be furnished as specified in 4.2.6.2.

(a) Test report of driving unit.
(b) Test report of pump.
(c) Test report on the combined assembled unit, including a definite report on the suitability of the driving unit (and controller, if furnished) for its specific application and harmony of assembly of all independent units going to make up the complete equipment.

4.2.6.2 The reports specified in 4.2.6.1 shall be distributed as follows:

(a) Three to the Bureau of Ships (to be forwarded via the cognizant inspector).
(b) One for primary district inspection office.
(c) One for branch inspection office (if any).
(d) Two to each supervisor of shipbuilding concerned (one for supervisor's files, the other for the shipbuilder).
(e) One to the Commander of each ship building yard concerned, for Navy yard built ships, when *specified in the contract or order.*

4.2.7 Shock tests.-

4.2.7.1 One complete pumping unit of each type, design and size shall be subjected to the high impact shock test of MIL-S-901 as specified in 3.2.2.

4.2.7.2 All pump units shall be shock tested with drivers unless otherwise approved by the bureau or agency concerned. Flexible coupled pumps shock tested with one driver will not be required to be shock

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tested again when supplied with a different driver. Prime movers will be subject to shock tests in accordance with the applicable equipment specifications.

4.2.7.3 Compliance with shock resistance requirements may be covered by test of a similar unit subject to bureau or agency concerned approval.

4.2.7.4 Equipment previously shock tested and accepted will not be required to be retested except when evidence of low shock resistance develops in the units installed. The bureau or agency concerned will initiate action with the manufacturer for correction of deficiencies, and may require shock tests at any time.

4.2.7.5 The pump unit shall be considered to have failed to pass the shock tests in the event of any of the following:

(a) Breakage of any parts, including mounting bolts.
(b) Appreciable distortion or dislocation of any part, such as shaft, mounting feet, bearings.
(c) A mechanical unbalance of more than two times the amplitude of unbalance measured prior to tests at rated speed, or more than that permitted by the specifications for the driver.

4.2.7.5.1 Pump units which have been subjected to the high-impact shock test and have failed to conform to the requirements of this specification will not be acceptable.

4.2.7.6 Pump units which have been subjected to the high-impact shock test and have successfully passed this test shall be considered acceptable for such service as the bureau or agency concerned may authorize. Such equipment shall be reconditioned by the manufacturer as follows:

(a) Minor deformations of mounting flanges shall be corrected.
(b) Minor deformations affecting alignment shall be corrected.
(c) All bearings shall be replaced.
(d) Each part shall be carefully examined by the contractor and any part which he considers substandard shall be replaced.

4.2.7.6.1 Minor deformations shall be defined as those which do not cause unqualified rejection of the design under high-impact shock test but which are in excess of the design dimensional tolerances specified on the applicable pump drawings.

4.2.8 Inspection and test of joints in systems designated as hazardous in NAVSHIPS 250-648-8 shall be in accordance with the requirements specified therein.

4.2.9 Hydrostatic tests.-

4.2.9.1 Pump cylinders and valve chambers shall be tested hydrostatically to one and one-half times the rated pump discharge pressure.

4.2.9.2 Steam cylinders and valve chests shall be tested hydrostatically to one and one-half times the maximum designed boiler steam drum pressure.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

5.1.1 Basic equipment.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation. Preservation and packaging may conform to the supplier's commercial practice when such meets these requirements. Manuals furnished with each pump unit (see 3.7) shall be packaged in sealed greaseproof, waterproof bags or containers.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers

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or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, class contract or order number, contractor's name and destination.

5.1.1.3.1 Special marking.- In addition to the marking specified in 5.1.1.3 each shipping container shall be marked "SERVICE MANUALS ENCLOSED".

5.1.2 Onboard repair parts.- Onboard repair parts shall be preserved and packaged Level A; packed Level B or C as specified (see 6.1) and marked in accordance with the requirements of the following specifications as applicable:

Mechanical parts	MIL-R-196 and MIL-P-17286
Electrical parts	MIL-E-17555
Tools	MIL-H-15424 and MIL-P-17286

5.1.2.1 Repair parts boxes.- When specified (see 6.1) repair parts shall be furnished in repair parts boxes conforming to type M or type W of MIL-B-233 as specified (see 6.1). When type W boxes are required they shall be fitted with non-magnetic hardware throughout.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation, packaging, packing and marking.-

5.2.1.1.1 Pump unit.- Each pump unit (see 1.1 and 1.2) and accessories shall be cleaned, preserved and packaged for domestic shipment-immediate use or domestic shipment and storage or for overseas shipment as specified; packed for domestic shipment and storage or overseas shipment as specified and marked in accordance with MIL-P-16789.

5.2.1.1.2 Repair parts and tools.- Repair parts and tools shall be preserved and packaged Level A or C as specified; packed Level A or B as specified and marked in accordance with the requirements of MIL-R-196, MIL-P-17286, MIL-E-17555 and MIL-H-15424, as applicable.

5.2.1.1.3 Service manuals.- Service manuals shall be prepared for shipment in accordance with MIL-M-15071 except, manuals furnished with each pump unit (see 3.7.5) shall be packaged in greaseproof, waterproof sealed bags or containers.

5.2.1.1.4 Drawings.- Drawings shall be prepared for shipment in accordance with MIL-D-963. Marking of interior packages and exterior shipping containers shall be in accordance with MIL-STD-129).

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Pump mounting required (see 3.2.5).
(c) If air chambers are required (see 3.2.19).
(d) If additional manuals are required (see 3.7.5).
(e) Repair parts required (see 3.8.3).
(f) Preservation, packaging, packing and marking required if other than as specified in 5.1 (see 5.2).

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Notice. - When Government drawings, specifications, or other data are used for any purpose other than in accordance with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way specified the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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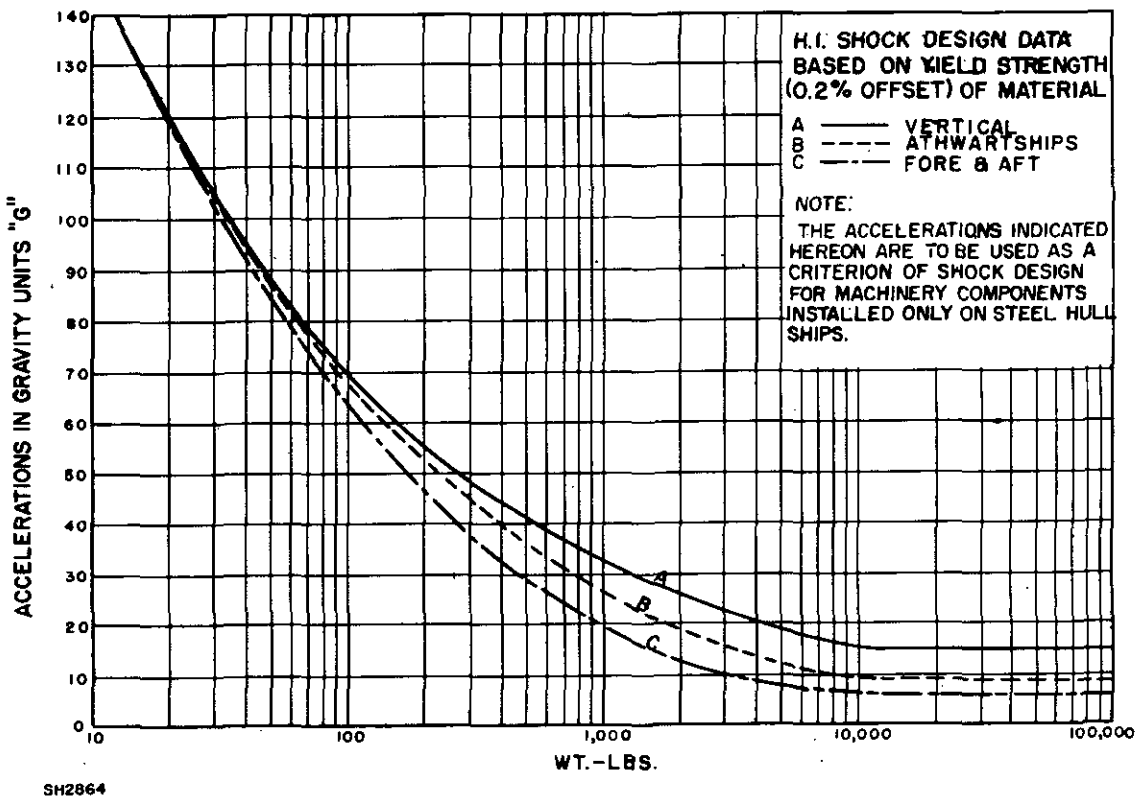


Figure 1.-H.I. Shock design data.

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

BUDGET BU. NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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